Household Stages

Axis-Factored Markov Kernels for Discrete-Time Heterogeneous-Agent Models

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July 4, 2026

Abstract

The household block of a discrete-time heterogeneous-agent model can often be decomposed as a sequence of within-period *stages*. I define and efficiently implement household stages as axis-factored (reward-tagged) Markov kernels. "Axis-factored" here means that stages act on one axis of a multidimensional idiosyncratic state space, while remaining flexibly dependent on the values along other axes. I show that the finite closure, under composition and direct sum, of a small set of *primitive* stages spans a large share of household blocks in the literature. The backward (value) and forward (distribution) operators of most of these primitive stages obey an adjoint property—the discrete-time analogue of the HJB–KFE adjointness of continuous-time models—that follows from the envelope theorem. This "envelope-adjointness" is preserved under composition and direct sum, and underlies the fake-news / sequence-space-Jacobian algorithm. Furthermore, it allows for efficient automatic differentiation (AD) of stages. I implement primitive stages, the two compositors composition and direct sum, and AD utilities in a Julia package, `HouseholdStages.jl`. I use language models to: (1) translate a large number of published papers' household blocks into the compositional stages algebra provided by this package, and (2) produce Appendix A listing 100 models whose household blocks can be built in this manner.

Note: This note is at present more or less a stub. I am currently working on expanding it. However, if you do use this package, or build on its methodology, please cite this as a working paper.

1 Stage Definition

Definition 1 (Stage)**.** *A* stage *is a two-way mapping between two idiosyncratic household state spaces* X_{start} *and* X_{end}*, consisting of two operators,*

$$T: \ \mathcal{M}(X_{\text{end}}) \to \mathcal{M}(X_{\text{start}}),$$

$$K: \ \mathcal{M}^*(X_{\text{start}}) \to \mathcal{M}^*(X_{\text{end}}),$$

the ***backward*** *value operator* $V_{\text{start}} = T\, V_{\text{end}}$ *and the* ***forward*** *simulation operator* $\Lambda_{\text{end}} = K\, \Lambda_{\text{start}}$.

Stages for which,

$$K = \left(D_{V_{\text{end}}} T\right)^*, \tag{1}$$

are said to obey the ***envelope-adjoint*** *property.*

2 Primitive stages

The primitives are of three kinds: (1) a *choice* family containing argmax, softmax (a.k.a. logit, taste shock discrete choice), and exogenous Markov transitions; (2) *affine shifts*; and (3) *pointwise scalings*. Each stage acts upon exactly one axis of household heterogeneity, though its parameters remain flexibly dependent on the values of other axes. Table 1 lists the set of primitive stages implemented by `HouseholdStages.jl`.

Closure. Stages are closed under two forms of composition. First, time-ordered *composition* $\circ$. Second, *direct sum* $\oplus$, which applies its component stages in parallel. A *household stage* is any element of the finite span of primitive stages under $\circ$ and $\oplus$.

Primitive	**Backward** $V_{\text{start}} = TV_{\text{end}}$	**Forward** $\Lambda_{\text{end}} = K\Lambda_{\text{start}}$	**Role**
`MarkovStage`	$T\,V_{\text{end}}$ (expectation)	push by $T^\top$	exogenous transition
`ArgmaxStage`	$\max_a[\,r_a + V_{\text{end}}\,]$ (tropical)	scatter by $\arg\max$	hard choice ($\varepsilon \to 0$)
`LogitChoiceStage`	$\varepsilon \log \sum e^{(-C+V_{\text{end}})/\varepsilon}$	push by logit prob.	soft choice (temperature)
`ContinuousArgmaxStage`	$\max_{a'}[\,r(a') + \text{interp}\,V_{\text{end}}\,]$	Young-split by $a'^\star$	off-grid choice (differentiable)
`DeterministicContinuousStage`	interp-gather at $d(\cdot)$	Young-split scatter to $d(\cdot)$	deterministic move (off-grid)
`DiscreteMoveStage`	gather at nearest index	0/1 scatter	deterministic move (snapped)
`UtilityStage`	$u + V_{\text{end}}$	identity	affine shift (backward)
`EntryStage`	identity	$\Lambda_{\text{start}} + g$	affine source (forward)
`PointwiseScaleStage`	$a \cdot V_{\text{end}}$	$f \cdot \Lambda_{\text{start}}$	diagonal scale ($a = f$: adjoint)
`TimeDiscountingStage`	$\beta \cdot V_{\text{end}}$	identity	discount ($a = \beta, f = 1$)
`IdentityStage`	V_{end}	Λ_{start}	no-op / $\oplus$ branch

Table 1: The primitive stages implemented by `HouseholdStages.jl`.

A Model catalogue

The 100 models below ship as runnable examples; each household block is a composition of the primitives of Table 1 that adds no bespoke stage (custom code lives in the surrounding equilibrium / driver; a few intricate cases use the auxiliary-choice-axis pattern). Models are organized by the within-period control mechanism. See the package's `MODEL_CATALOG.md` for the per-model stage chains and annotations.

§0 Spine — exogenous dynamics, level shifts, savings core. aiyagari, huggett, bewley, imrohoroglu, deaton, carroll_buffer_stock, castaneda, krueger_mitman_perri, kaplan_violante, income_ar1, persistent_transitory, countercyclical_risk, regime_switching, heterogeneous_income, diagnostic_expectations, hsv_progressive_tax, unemployment_insurance, borrowing_constraint, taste_shocks, discount_heterogeneity, money_in_utility, cash_in_advance, limited_commitment, aiyagari_mit_shock, auclert_revaluation, krusell_smith, life_cycle.

§1 Drift / level control — continuous "how much". human_capital, health, capital_investment, rnd_investment, marriage_capital, guvenen_kuruscu, manuelli_seshadri, hall_jones, becker_tomes, wealth_in_utility, labor_supply, indivisible_labor, collective_household, temptation.

§2 Dispersion / variance control — RI, portfolio, risk, robustness. portfolio, cocco_gomes_maenhout, catherine, gomes_michaelides, fagereng, cara_normal, uninsured_investment_risk, rational_inattention, mackowiak_wiederholt, hansen_sargent, ilut_schneider, risk_shifting, entrepreneurship, discrete_ri.

§3 Hazard / probability control — search, effort, survival, attention. search_matching, mccall_search, endogenous_separations, skill_depreciation, promotion_ladder, longevity_effort, soft_default, insurance.

§4 Direction / support control — migration, matching, occupation. spatial, directed_search, sectoral, choo_siow, monte_redding_rrh, bryan_morten, tombe_zhu, diamond, hsieh_hurst_jones_klenow, bayer_ferreira_mcmillan, technology_adoption, menzio_shi.

§5 Discrete jumps / stopping; composites; demographics. default, durable_housing, habit, two_asset_hank, two_asset_hank_pv, arellano_default, livshits_macgee_tertilt, rust_replacement, vintage_replacement, mortgage_refinancing, constantinides_habit, durable_liquid, perpetual_youth, conesa_krueger, de_nardi_bequests, de_nardi_french_jones, entry_birth.

§6 Firm dynamics (same stages, reinterpreted). hopenhayn, clementi_palazzo, lumpy_investment, lumpy_labor, menu_cost, inventory_ss, collateral_investment.

§7 Sign-flips. costly_diffusion (the "diffuse" dual of rational inattention); the remaining flips—robustness as $\varepsilon < 0$, gambling as convex V, entry$\leftrightarrow$exit—are realized by models already listed above.